

BioHackathon series:

[SWAT4HCLS Biohackathon 2026](#)

Amsterdam, The Netherlands

[Project 5](#)

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SWAT4HCLS Biohackathon 2026: Template for the very long title

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SWAT4HCLS Biohackathon 2026: Getting LLM Agents on the Same Page

Introduction

As part of the SWAT4HCLS Hackathon 2026, we report on Project 5. This project explores the use of Large Language Model (LLM)-based agents for automated reconciliation of domain-specific terminology with structured ontology concepts. Semantic interoperability remains a major bottleneck in achieving FAIR data integration across biomedical and life sciences domains. Mapping heterogeneous terminology to standardized identifiers from resources such as Wikidata and BioPortal is essential but typically relies on manual curation. This is a time-consuming and difficult to scale process. Recent advances in LLM-based agent systems provide new opportunities to automate this process by combining semantic similarity, contextual reasoning, and tool-based API interaction. In this project, we evaluate a multi-agent system designed to perform ontology mapping and apply it to real-world biomedical data derived from the [Synodos NF2 project](#).

Methods

System architecture

We implemented a multi-agent pipeline for automated mapping of domain-specific terms to ontology concepts. The system integrates:

- Ontology recommendation and search
- Definition retrieval and enrichment
- LLM-based reasoning for semantic alignment
- Assignment of structured identifiers and SKOS relations

The architecture is implemented using LangChain, LangGraph, and a Deep Agent framework.

Multi-agent design

The system consists of three coordinated agents: - Ontology Agent - BioPortal Agent - Orchestrating Agent

Ontology Agent The ontology agent selects relevant ontologies using the BioPortal recommender services.

BioPortal Agent The Bioportal agent is responsible for retrieving candidate concepts. This includes labels, synonyms, and definitions.

Orchestrating Agent The Orchestrating agent integrates results, evaluates candidates, and assigns: - Best matching identifier (Wikidata QID or ontology URI) - SKOS semantic relation (e.g., exactMatch, closeMatch) - Explanation for the mapping

Data source: Synodos NF2 dataset

We evaluated the system using a subset of data from the Synodos NF2 Drug Screening Dataset, hosted on Synapse.

The dataset can be found accessed through the DOI down below.

- DOI: <https://doi.org/10.7303/syn6138237.1>
- File: Synodos_DrugScreen_processed_data.tsv

Additional related raw datasets (single-agent and combination screening) were also provided and used as supporting sources of terminology.

Case study: NF Data Portal

In the US funding agencies often require projects to upload experimental data to public data portals. This data sharing however, is often done after publication and thus user-submitted data critically lacks standardization and annotation. As a result while biomedical data portals, often contain large amounts of experimental data findability and usability are large scale bottlenecks limiting the re-use of this data in practice.

One such data portal [The NF Data Portal](#) developed by [Sage Bionetworks](#) is a large scale-repository for Neurofibromatosis-type disease studies. While the current NF Data Portal Web-UI, does provide study level metadata, it crucially does not provide grounded annotations of compounds explored in drug-screening studies. As such cross-study drug analysis on the portal currently requires users to manually inspect individual data files. To alleviate this bottle-neck we have applied our method to annotate multiple drug-screening studies from the NF Data Portal.

Meeting information

`biohackathon_name: "SWAT4HCLS Biohackathon 2026"`

`biohackathon_url: "https://www.swat4ls.org/swat4hcls-biohackathon-2026/"`

`biohackathon_location: "Amsterdam, The Netherlands"`

`group: Project 5`

`git_url: "https://github.com/AJKellmann/SWAT4HCLS26_hack-5-llm-agents-term-reconci"`

Citation Typing Ontology annotation

You can use [CiTO](#) annotations, as explained in [this BioHackathon Europe 2021 write up](#) and [this CiTO Pilot](#). Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethod In:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- `citesAsDataSource`: when you point the reader to a source of data which may explain a claim
- `usesDataFrom`: when you reuse somehow (and elaborate on) the data in the cited entity
- `usesMethodIn`
- `citesAsAuthority`

- citesAsEvidence
- citesAsPotentialSolution
- citesAsRecommendedReading
- citesAsRelated
- citesAsSourceDocument
- citesForInformation
- confirms
- documents
- providesDataFor
- obtainsSupportFrom
- discusses
- extends
- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

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Acknowledgements

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References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, 32(14), 2236–2238. <https://doi.org/10.1093/bioinformatics/btw214> [cito:citesAsAuthority]

Author contributions

Iurii Savvateev: Software, validation Woodward Galbraith: Software, validation Linda Hendriks: Writing – original draft, validation Alexander Kellmann: Writing – original draft, validation Nalini Paijens: Writing – original draft Umit Sude Böhler: Writing – original draft, validation